CASE 04 · ADVANCED BUSINESS ANALYST PROJECT · HEALTHCARE

The $131K Case for Earlier Screening

A predictive risk-screening model — Logistic Regression vs. Random Forest, on real clinical data.

Role simulated: Senior Business Analyst / Care Management Analytics | Tools: Python (scikit-learn — classification, ROC/AUC, feature importance)

Source: Pima Indians Diabetes Database (National Institute of Diabetes and Digestive and Kidney Diseases, public) — 768 patients

0.82	70%	35%	$131K
Model AUC (Logistic Regression)	Of at-risk patients caught (recall)	Base rate of diabetes in the data	Illustrative net benefit, test-set-scale outreach

The business problem

A payer or provider wants to run a proactive outreach program for patients at elevated diabetes risk — but nurse-educator capacity is limited, so outreach needs to be targeted at whoever a model can identify as actually at risk, not blanket-mailed to an entire population. This project builds and evaluates that targeting model, then translates its accuracy into a staffing-relevant business case.

Population note, stated plainly: this dataset covers only female patients of Pima Indian heritage, age 21+, near Phoenix, Arizona. A deployed model needs revalidation on the population it will actually screen — the methodology transfers, the specific thresholds may not.

Data preparation

Five fields can't biologically be zero (glucose, blood pressure, skin thickness, insulin, BMI) but the raw data encodes missing readings as 0 — a data-quality step most junior analyses skip. Recoded as missing and median-imputed: 227 missing skin-thickness readings, 374 missing insulin readings, 35 missing blood pressure, 11 missing BMI, 5 missing glucose. Insulin alone was missing for roughly half the dataset — worth flagging to a stakeholder before quoting model accuracy off it.

Model comparison

Both models were trained on a 576-patient training split and evaluated on the same held-out 192 patients. Random Forest edges out on raw AUC (0.827 vs. 0.823) but ties on recall (70%) — the metric that matters most for an outreach program, where missing an at-risk patient is costlier than one extra unnecessary call. Given that tie, Logistic Regression is the better production choice: its coefficients are directly explainable to a clinical stakeholder as a risk scorecard, where a Random Forest is a black box.

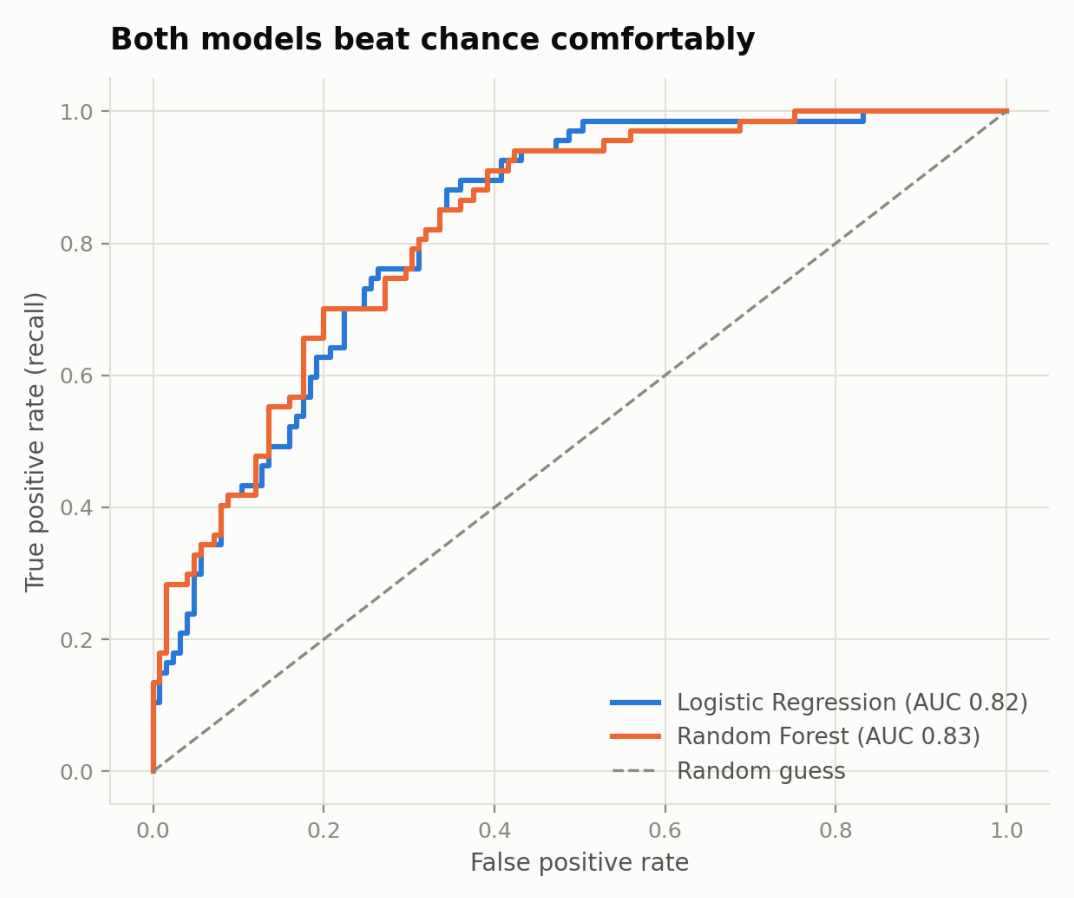


Fig. 1 — ROC curves, both models vs. random chance.

Metric	Logistic Regression	Random Forest
Accuracy	75.0%	72.9%
Precision	62.7%	59.5%
Recall (sensitivity)	70.1%	70.1%
F1 score	0.662	0.644
ROC-AUC	0.823	0.827

What's actually driving risk

Random Forest's feature importances (used here for interpretability of the underlying signal, independent of which model ships to production) point to the same clinical intuition a physician would expect — which is itself a useful model-validation check before trusting either model's predictions.

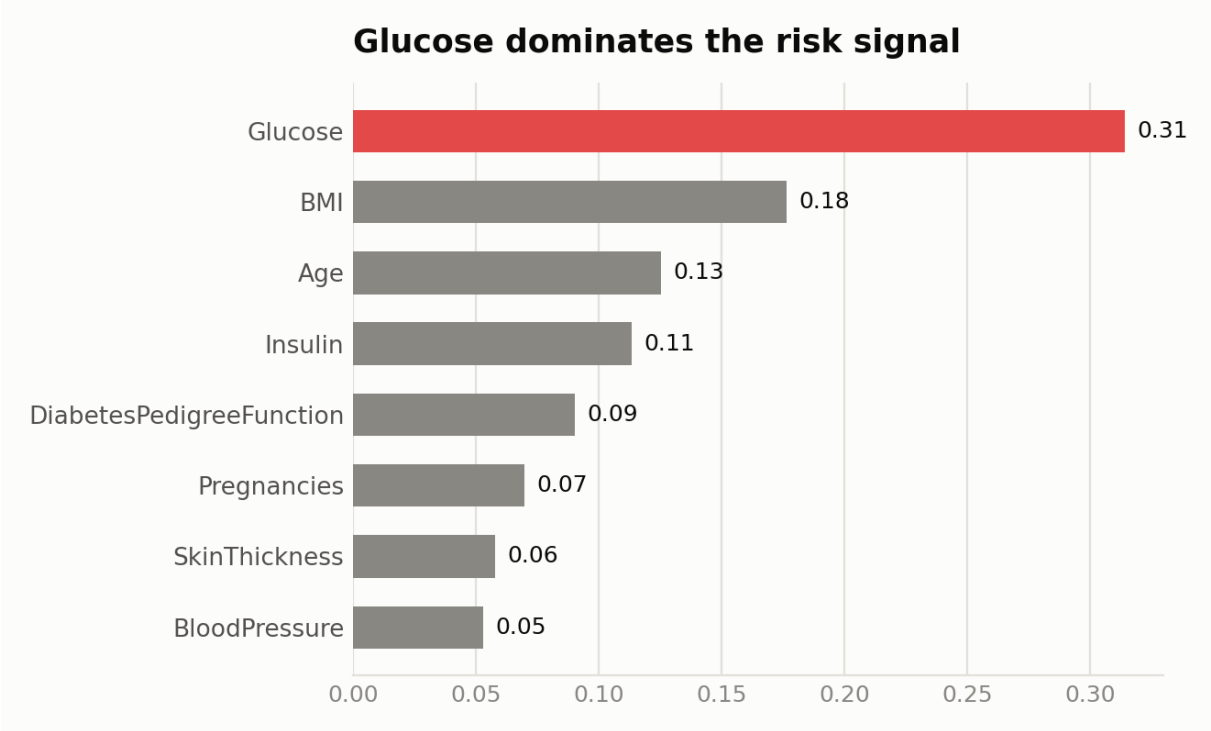


Fig. 2 — Random Forest feature importances.

Confusion matrix & the outreach math

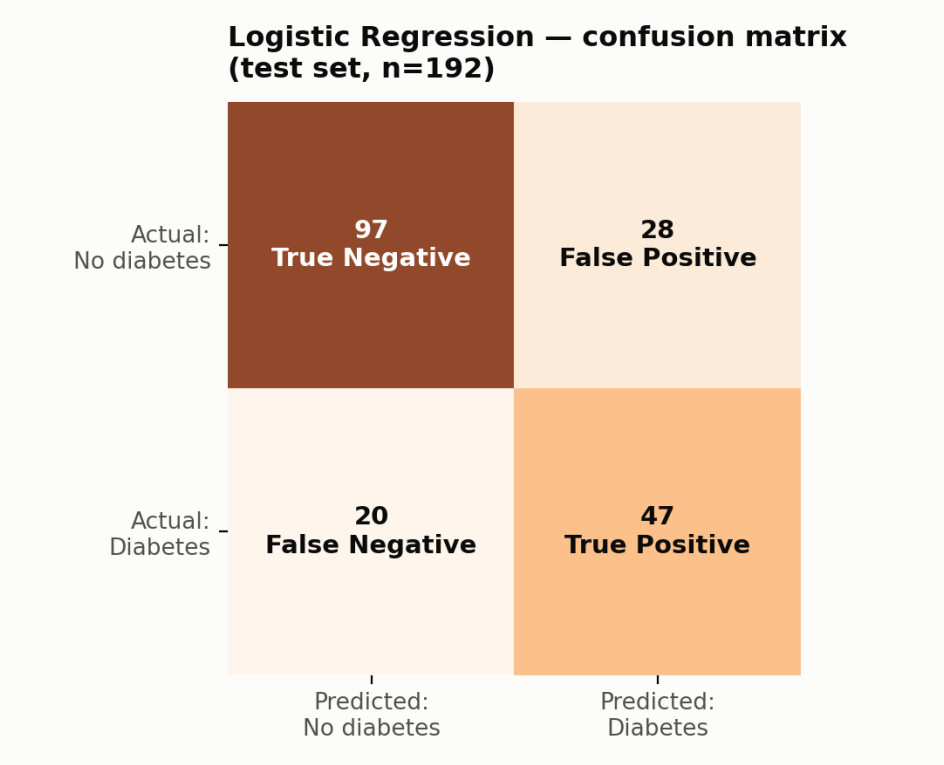


Fig. 3 — Logistic Regression confusion matrix on the 192-patient test set.

Scaled to this test set: screening 192 patients at an assumed $50/patient outreach cost (nurse call + materials — a placeholder, replace with your program's real cost) totals $9,600. The model correctly flags 47 of 67 actually-at-risk patients. Crediting only those true positives — a deliberately conservative assumption — at $3,000 of avoided downstream cost each (illustrative planning figure, not a specific cited statistic) yields $141,000 in avoided cost against the $9,600 program spend: a net benefit of $131,400. The takeaway to defend in an interview isn't the exact dollar figure — it's the shape of the argument: early identification is high-leverage precisely because untreated complications are so much more expensive than a phone call. Nationally, diagnosed diabetes costs the U.S. $412.9B annually and people with it have medical expenditures 2.6x higher than they would otherwise (ADA, 2022 data) — the direction of this case's math is well supported even where the specific placeholder dollar figures are not.

Recommendation

Deploy the Logistic Regression model as a risk-tiering scorecard (e.g., top-quartile predicted probability = high-priority outreach), not a diagnosis tool — its job is to prioritize limited nurse-educator capacity, and a 70% recall at a defensible, explainable threshold beats an un-targeted or purely intuition-driven outreach list.

RESUME BULLET, READY TO USE

“Built and compared Logistic Regression and Random Forest classifiers on 768 real patient records (AUC 0.82), selecting the interpretable model for production; identified glucose, BMI, and age as the dominant risk drivers and translated model recall into a care-management outreach business case.”

What's in this package

File	What it is
run_analysis.py	Data cleaning, both models, evaluation metrics, and all charts — fully reproducible.
model_results.json	Every metric behind this report, in machine-readable form.
This report (PDF)	Data-quality notes, model comparison, feature drivers, and the outreach business case.